



Mini review

A probabilistic and dynamic reformulation of expected goals (xG): Methodological advances and modelling perspectives

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ABSTRACT

Expected Goals (xG) is a probabilistic metric that estimates the likelihood of a shot resulting in a goal and is based on factors such as shot location, angle, body part used, and match context. It has become highly relevant in football analysis and is widely used to evaluate team performance, assess individual players, and guide recruitment and tactical decisions. In this study, we review the limitations of xG and propose strategies to improve its accuracy and applicability. Despite its popularity, xG is often misinterpreted as a deterministic measure. An xG value of 1 does not guarantee a goal but represents the expected outcome on the basis of shot probabilities. Summing shot-level xG values can be misleading, as multiple low-probability shots are not equivalent to a single high-probability attempt. Treating each shot as an independent probabilistic event, for example, using Bernoulli trials and Poisson binomial distributions, captures football's inherent uncertainty more accurately. Matches are dynamic, with teams adapting strategies throughout; thus, xG must be interpreted in context. Goals may occur outside traditional xG events, such as deflections or self-goals, and the low-scoring nature of football highlights that xG alone may not fully reflect performance. Player- and position-specific variations, along with situational factors such as match location, scoreline, and opponent strength, further affect outcomes. Moving from simple metrics to probabilistic modelling, future developments could integrate richer datasets, contextual variables, and real-time updates, transforming xG into a context-aware, actionable decision-support tool for coaches and analysts.

1. Expected goals (xG): concept, development, and applications

The expected goals (xG) metric has become a well-established tool within the football analytics community [1–3], although the volume of peer-reviewed academic research on the topic remains relatively limited. A considerable portion of research has been developed in grey literature, including master's theses and conference proceedings [4–6].

The conceptual origin of xG can be traced back to 2012, when Sam Green [7] proposed assigning a probability to each goal-scoring opportunity. Since then, the metric has evolved significantly, driven largely by private companies such as Opta Sports and professional football clubs. Despite this progress, much of the academic literature has focused on predicting match outcomes—often from a betting perspective

[8–10]—rather than on validating the reliability and robustness of the xG models themselves.

From a methodological perspective, xG models are typically built using machine learning techniques and large datasets of historical shots [1,2]. These models incorporate multiple input variables, such as distance and angle to goal, body part used, type of play (open play, penalty, free kick, or corner), defensive pressure, and match context. By training on extensive shot data, the model estimates the probability that a given attempt will result in a goal. Importantly, match-level xG is calculated as the sum of the probabilities of individual shots, making it a continuous measure, whereas the actual number of goals scored is a discrete outcome.

In professional football, xG has three main applications [1,2,11–14]. First, it is widely used to evaluate team performance. Since football is a

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low-scoring sport in which randomness can strongly influence results, xG provides insight beyond the final scoreline by estimating which team created better scoring opportunities and was more likely to win. Second, xG can be applied to individual player analysis. Player-specific xG and expected assists (xA) values allow analysts to assess a player's contribution to goal-scoring opportunities, finishing efficiency relative to expectations, and the ability to create chances for teammates. Third, xG is increasingly used in recruitment and scouting, helping clubs identify potential transfer targets whose underlying performance may not be fully reflected in observed outcomes.

Complementary metrics, such as Post-Shot Expected Goals (PSxG) and Expected Goals on Target (xGOT), have further extended the analytical framework. PSxG focuses on the quality of execution after the shot is taken, incorporating factors such as trajectory, speed, and goalkeeper positioning, thereby providing a more refined estimate of scoring probability. Similarly, xGOT and the analysis of off-target attempts offer additional insights into offensive intent and shot selection [15,16]. However, these extensions are often treated as complementary tools rather than being fully integrated into a unified probabilistic framework, limiting their contribution to a comprehensive evaluation of match performance.

The xG metric has been widely used to estimate match outcomes and final standings in football competitions and has shown significant predictive power [1,2,15–25]. [15,16,27,28]. In particular, compared with xG alone, PSxG often demonstrates greater predictive power, highlighting the importance of shot execution quality. In practical terms, these metrics allow coaches to set tactical thresholds; for instance, a + 1.0 PSxG advantage may correspond to an 85% probability of winning. They can also support player evaluation and scouting by quantifying the impact of individual performances on expected outcomes [16,29,30]. Across different xG differences, the probability of losing systematically decreases, whereas the probability of winning increases. At the same time, the likelihood of a draw peaks when the difference in xG is close to zero. Together, these findings show how xG-based metrics help explain match results beyond the final scoreline.

2. Expected goals in football: from simple metrics to probabilistic modelling

The xG metric has been rapidly adopted across multiple football contexts, ranging from fans to professional coaches. However, despite its widespread use, its interpretation, both in terms of on-field performance and its mathematical formulation, may not always be fully accurate. xG is frequently treated as a deterministic indicator of performance rather than as a probabilistic estimate [1,18,26,28]. An xG value of 1 at the end of a match does not imply that the team was guaranteed to score a goal. Rather, it reflects that, based on the cumulative probability of all shots taken during the match, the expected outcome was approximately 1 goal. Having a high or low probability of achieving a given outcome does not mean that less likely outcomes cannot occur. In other words, what is probable is not the same as what is guaranteed. This distinction is fundamental. Such misunderstandings have likely contributed to the misinterpretation of xG and its explanatory potential, leading in practice to an overreliance on the metric in applied analyses and media commentary. Consequently, it is essential to interpret xG in probabilistic terms rather than as an absolute or deterministic measure.

Building on this caution, the simple summation of shot-level xG values and their use as a definitive measure of match performance can be misleading. For example, five shots with an xG value of 0.20 are not equivalent to a single shot with an xG value of 1.00, neither statistically nor in terms of their meaning within the game. Therefore, this issue should be approached with care. Aggregating individual xG values from each shot and assuming that this provides a clear representation of team performance in a match can be problematic and potentially misleading. Accordingly, the assessment of what occurred in a match should rely solely on the final aggregated xG value derived from all the shots. In

essence, the sum of individual xG values should not be uncritically interpreted as a final performance metric. Each shot should be treated as an independent observation, with its own associated probability [28,29].

Expanding on this idea, treating each shot and its xG value as an independent observation can provide a much more nuanced understanding of match dynamics. Recently, it has been proposed to model each shot as an independent Bernoulli trial within a Bayesian probabilistic framework [28]. This approach accounts for the fact that not all shots are equal, as some have higher xG values because of distance, angle, or other contextual factors; thus, the probability distribution of total goals cannot be accurately captured by a simple linear sum. Consequently, using the Poisson binomial distribution, it is possible to calculate the exact probability of scoring any number of goals and, consequently, the probabilities of winning, drawing, or losing a match. For instance, in a 2024–2025 top-tier professional football match between two elite teams, a total of 24 shots were recorded, 13 for Team A and 11 for Team B. When these shots were treated as independent Bernoulli events, the probabilities of scoring were approximately as follows: For Team A, 31 % for 0 goals, 40 % for 1 goal, and 29 % for 2 or more goals; for Team B, 38 % for 0 goals, 38 % for 1 goal, and 24 % for 2 or more goals. From these distributions, the predicted match outcomes were as follows: Team A win 65.5 per cent, draw 20.1 per cent, and lose 14.4 per cent; Team B win 14.4 per cent, draw 20.1 per cent, and lose 65.5 per cent (see Table 1) [28]. This example illustrates how modelling each shot as an independent probabilistic event and using the Poisson binomial distribution captures the inherent uncertainty of football, producing a more accurate and nuanced representation of match dynamics than simply summing xG values linearly.

Although the present work illustrates the approach using a single high-profile match, we believe that the proposed probabilistic framework is equally applicable to lower-scoring matches, different leagues, and women's football. Since the method is based on modelling each shot as an independent probabilistic event, its underlying logic is not restricted to a specific competition or match context. Therefore, the same approach can be extended to different football environments, although future studies should further evaluate its robustness and potential contextual differences across competitions and playing levels.

Moving from methodology to match dynamics, it is important to understand that football matches are constantly evolving and that teams can adjust their style of play or performance as the match unfolds. As a result, each match can contain several phases. Therefore, performance analysis based on the xG metric should be more dynamic and adapted to the changing reality of each game. To enable practical applications and help coaches make better decisions, it is necessary to recalculate the probabilities of winning, drawing, or losing immediately after key events, as well as segment matches into different phases to provide information that truly helps in understanding what is happening in the game. By analysing the match in phases before and after key events such as the first goal, a player substitution, a sending-off, or a change in the coach's tactical plan, changes in tactical and behavioural patterns that whole-match analyses might obscure can be captured [30–32]. This sequential perspective allows coaches and analysts to identify critical moments that influence outcomes, understand which tactical decisions were effective, and evaluate the impact of individual players on team performance [26,30]. From a methodological point of

Table 1
Probabilities of match outcomes in a match between two elite professional football teams derived from xG values.

Match Outcome	PrexG	95% CI
Team A wins	14.40%	8.85–22.71
Draw	20.09%	13.49–28.99
Team B wins	65.51%	55.88–74.24

view, this shift in approach relies on modelling each shot as an independent Bernoulli trial, where the probability of success corresponds to the xG value of that shot [28]. The sum of these trials can be modelled using the Poisson binomial distribution, which allows calculation of the probability of scoring 0, 1, 2, or more goals for each team without assuming linearity or homogeneity of the shots. Modelling each shot as an independent probabilistic event captures the inherent uncertainty of football and reflects the evolving dynamics of the game, offering a far more accurate and useful representation of match performance than simply summing xG values linearly. Furthermore, by integrating these calculations within a Bayesian framework, it is possible to continuously update the probabilities of partial and final match outcomes as events unfold. As a result, analysts can dynamically recalculate win, draw, or loss probabilities after major events, and segment matches into tactical phases during video review, providing actionable insights for coaching, tactical adjustments, and player evaluation.

While highlighting the current structural limitations of the xG metric, it is also crucial to improve its predictive accuracy. For instance, a team can score a goal without taking any shots that register as xG. This creates seemingly counterintuitive situations, such as an xG of 0 despite scoring a goal. Events such as own goals, crosses deflected by an opposing player, or goalkeeper errors that do not result from a formal shot are not captured by the metric, even though they can determine the outcome of a match. Therefore, the metric must continue to be refined to provide a more precise understanding of both the final score and the evolution of partial match outcomes.

Another aspect that warrants attention is that the xG metric captures only part of the team's performance during a match. In low-scoring sports such as football, shots on goal are relatively rare events. For example, in LaLiga during the 2025–26 season, teams averaged 13.5 total shot attempts per match, of which only 4.3 were on target. This scarcity suggests that relying solely on xG may provide an incomplete view of match dynamics. Therefore, combining xG with additional metrics can offer a broader perspective on what is happening on the pitch. Metrics such as Expected Possession Value (EPV), and Expected Ball Gain (EPG) can be particularly useful in this regard [3,33]. EPV measures the value of an action while a team maintains possession, estimating the likelihood that a possession sequence will lead to a goal based on ball location and player decisions. Moreover, EPG quantifies the expected value of regaining possession and evaluates how defensive actions, such as interceptions, tackles, or ball recoveries, increase the probability of generating a scoring opportunity for the team. Together, these metrics allow analysts and coaches to capture both offensive and defensive contributions that may not be fully reflected in xG alone, providing a more comprehensive understanding of team performance and match dynamics.

Situational factors play a central role in understanding football performance, as player and team output is strongly influenced by elements such as match location (home or away), match status (winning, drawing, or losing), and the quality of the opposition [30–32]. For example, teams may reduce high-intensity activity when leading and increase it when behind, home teams often adopt more offensive and dominant tactics, and weaker teams cover greater distances when facing stronger opponents. Considering these factors allows analysts and coaches to evaluate performance more accurately, adjusting both pre-match preparations and post-match assessments to reflect the true context of the game. In this framework, the xG metric becomes a particularly valuable tool, offering a probabilistic perspective on goal-scoring opportunities across different match scenarios. By integrating situational variables such as scoreline, match location, and opponent strength, xG allows for a more nuanced understanding of team and player performance. Moreover, post-match evaluations can be made more objective by incorporating these contextual factors, ensuring that interpretations of both individual and collective outcomes capture the dynamic reality of the game and the conditions under which it was played.

Another aspect in which the xG metric should improve is its predictive accuracy, through richer datasets, more advanced machine learning pipelines, and better identification of the factors that increase or reduce the probability that a shot results in a goal. Recently, several studies have suggested that there are position and player effects on xG, meaning that certain positions or players have higher or lower goal probabilities for a given chance than others [11,34]. For instance, strikers and attacking midfielders tend to exhibit positive xG adjustments, whereas defenders show, on average, negative adjustments. Similarly, imagine that two players take the same shot from the same position, with defenders located in identical positions and all other conditions being equal. One player represents an elite-level attacking profile with outstanding finishing efficiency, whereas the other corresponds to an average performer from a semiprofessional competition. Therefore, assuming comparable contextual conditions, the elite player would be expected to demonstrate a higher likelihood of scoring. However, the xG metric would assign the same value to both attempts under these circumstances. Therefore, further work is needed to refine the xG metric by incorporating additional factors that better capture a shot's potential success.

3. Conclusions and future directions for xG

In conclusion, the future development of the xG metric should focus on reinforcing its probabilistic interpretation and avoiding deterministic readings of expected values. Methodologically, moving beyond the simple aggregation of shot-level xG towards approaches that model each shot as an independent probabilistic event, such as within a Poisson binomial framework, would provide a more accurate representation of match uncertainty. Moreover, xG models should evolve to incorporate richer datasets and more advanced analytical techniques, including player- and position-specific effects, as well as contextual and situational variables that influence performance. In addition, adopting a more dynamic perspective, in which probabilities are updated across different phases of the match, would further enhance its practical applicability. Importantly, xG should be understood as a measure that moves beyond merely reflecting what happened and instead focuses on what could have happened, capturing the underlying potential of each scoring opportunity.

Looking ahead, these advances will make xG a truly actionable tool for coaches and analysts. By enabling real-time updates of goal probabilities and match outcome expectations, the metric can inform tactical decisions as the game unfolds, not just in post-match reviews. Moreover, integrating xG with complementary metrics and situational data will allow coaching staff to identify critical moments, adjust strategies dynamically, and evaluate player contributions more accurately. Ultimately, the future of xG lies in transforming it into a dynamic, context-aware decision-support system that bridges the gap between statistical analysis and practical, on-field decision-making.

CRedit authorship contribution statement

Lago Peñas Carlos: Writing – review & editing, Writing – original draft, Conceptualization. **Runze Feng:** Writing – original draft, Conceptualization. **Yeqin Zhang:** Writing – review & editing, Conceptualization. **Wuzhi Dong:** Writing – review & editing, Conceptualization. **Bingnan Gong:** Writing – original draft, Conceptualization. **Gómez-Ruano Miguel-Ángel:** Writing – original draft, Conceptualization.

Declaration of Competing Interest

Carlos Lago-Peñas is an Associate Editor, and Miguel-Ángel GómezRuano is an Editor-in-Chief for Intelligent Sports and Health. They were not involved in the editorial review or the decision to publish this article. No financial relationships have influenced the research presented in this manuscript.

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